

Program : Diploma in Instrumentation Engineering	
Course Code : 3086	Course Title: Sensors and Transducers Lab
Semester : 3	Credits: 1.5
Course Category: Program Core	
Periods per week: 3 (L:0 T:0 P:3)	Periods per semester: 45

Course Objectives:

- To be familiar with the use different transducers and sensors
- To make the students aware of the characteristics of these transducers.

Course Prerequisites:

Topic	Course code	Course name	Semester
Basic principle of resistance, inductance, capacitance, diode , transistors etc.		Fundamentals of Electrical & and electronics	2
Permeability, Electromagnetic induction, self and mutual inductance; Transformer.		Fundamentals of Electrical & and electronics.	2

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Plot the characteristics of resistive transducers and measure various parameters using resistive transducers	18	Applying
CO2	Measure various parameters using capacitive, inductive and piezo electric transducers and plot the characteristics of inductive transducers.	9	Applying
CO3	Plot the characteristics of photoelectric transducers and measure various parameters using ultra sonic and hall effect transducer	13	Applying
	Lab Test	5	

CO – PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	3	3	3			2
CO2	3	3	3		3		
CO3	3	3	3		3		3

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Plot the characteristics of resistive transducers and measure various parameters using resistive transducers		
M1.01	Measure of displacement using potentiometer.	3	Applying
M1.02	Measure the loading error in potentiometer.	3	Applying
M1.03	Plot the characteristics of strain gauge.	3	Understanding
M1.04	Plot the characteristics of LDR.	3	Understanding
M1.05	Plot the characteristics of Thermistor.	3	Understanding
M1.06	Plot the characteristics of RTD.	3	Understanding
Contents: Measure of displacement using potentiometer, Measure the loading error in potentiometer, Plot the characteristics of strain gauge, LDR, Thermistor, RTD.			
CO2	Measure various parameters using capacitive, inductive and piezo electric transducers and plot the characteristics of inductive transducers.		
M2.01	Plot the characteristics of LVDT.	3	Understanding
M2.02	Measure level/displacement using capacitive transducer.	3	Applying
M2.03	Measure dynamic pressure/vibration using piezo electric transducer.	3	Applying
	Lab Test – I	3	

CO3	Plot the characteristics of photoelectric transducers and measure various parameters using ultra sonic and hall effect transducer		
M3.01	Plot the characteristics of Photo diode.	3	Understanding
M3.02	Plot the characteristics of Photo transistor.	3	Understanding
M3.03	Measure displacement/ level using ultrasonic transducer.	4	Applying
M3.04	Detect magnetic flux using Hall Effect transducer.	3	Applying
	Lab Test – II	2	

Text / Reference:

T/R	Book Title / Author
R1	A.K. Sawhney, “ A course in Electrical and Electronic measurement and Instrumentation”, Dhanpathrayi and Co.
R2	D.V.S. Murthy, “ Transducers and Instrumentation”, PHI. Pvt. Ltd.
R3	D.Patranabis, “ Principles of Industrial Instrumentation”, Tata McGraw-Hill. Education,
R4	B. C. Nakra, K. K. Chaudhry, “Instrumentation, Measurement And Analysis”, Tata McGraw-Hill Publishers.
R5	Bela G. Liptak, “Process Measurement and Analysis”, Chilton Book Company.
R6	Ranganathan S, “ Transducer engineering ”, Allied publishers.
R7	R.K. JAIN, “Industrial Instrumentation”, Khanna Publishers.
R8	Krishna Kant, “Computer-Based Industrial Control”, PHI.
R9	D Patranabis, “ Sensors and Transducers”, PHI.
R10	Ernest O. Doebelin, “Doebelin's Measurement Systems”, McGraw-Hill.

Online Resources:

Sl.No	Website Link
1	https://www.youtube.com/watch?v=FYggY7LLDrw
2	https://www.slideshare.net/manash234/classification-of-transducers
3	https://www.electronics-tutorials.ws/io/io_1.html
4	https://www.egr.msu.edu/classes/ece445/mason/Files/4-Sensors_ch2.pdf
5	https://www.youtube.com/watch?v=XSbUnUADlno
6	http://www.kyowa-ei.com/eng/technical/strain_gages/principles.html

7	https://www.youtube.com/watch?v=anCnrtjNLQM
8	http://www.te.com/content/dam/te-com/documents/sensors/global/lvdt-tutorial.pdf
9	https://www.electronics-tutorials.ws/electromagnetism/hall-effect.html
10	https://www.brighthubengineering.com/hvac/53147-how-capacitive-transducers-work/
11	https://www.powershow.com/view4/556e94-ODlIO/TRANSDUCERS_VARIABLE_RESISTIVE_CAPACITIVE_INDUCTIVE_powerpoint_ppt_presentation
12	https://www.ctscorp.com/resource-center/tutorials/piezo-basics/
13	https://www.youtube.com/watch?v=CckMERCuXvE
14	https://www.youtube.com/watch?v=7jQ12uUqK6E